

Polymer Dynamics

Blockwise MSD and Radius of Gyration Analysis

Weekly Progress Report 3

Submitted by

Nida Farheen C M

Submitted to

Dr. Debarati Chatterjee

Indian Institute of Technology Palakkad

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1 Introduction

Polymer dynamics during thermal annealing were analyzed using Molecular Dynamics trajectories generated from LAMMPS.

The study focuses on understanding how polymer mobility and structural properties evolve across different temperature windows during annealing.

Two major quantities were analyzed:

- Mean Square Displacement (MSD)
- Radius of Gyration (R_g)

The trajectory was divided into multiple timestep blocks corresponding to different annealing stages, and all quantities were computed independently within each block.

2 Objectives

1. Compute blockwise Mean Square Displacement (MSD)
2. Analyze subdiffusive polymer dynamics
3. Fit MSD using power-law scaling
4. Compute Radius of Gyration (R_g)
5. Study structural evolution during annealing
6. Visualize temperature-dependent polymer behavior

3 Simulation Details

The supplied LAMMPS trajectory contains polymer monomer coordinates stored using unwrapped coordinates:

$$xu, yu, zu$$

The trajectory was divided into seven annealing blocks:

Block	Timestep Range
1	$0 - 2.5 \times 10^7$
2	$2.5 \times 10^7 - 5 \times 10^7$
3	$5 \times 10^7 - 7.5 \times 10^7$
4	$7.5 \times 10^7 - 1.0 \times 10^8$
5	$1.0 \times 10^8 - 1.25 \times 10^8$
6	$1.25 \times 10^8 - 1.5 \times 10^8$
7	$1.5 \times 10^8 - 1.75 \times 10^8$

Table 1: Temperature annealing blocks used for analysis

4 Mean Square Displacement (MSD)

The Mean Square Displacement measures the average displacement of polymer monomers over time and is defined as:

$$MSD(\tau) = \langle |\mathbf{r}(t + \tau) - \mathbf{r}(t)|^2 \rangle$$

where:

- $\mathbf{r}(t)$ is the monomer position at time t
- τ is the lag time
- averaging is performed over all valid time origins and monomers

An FFT-based algorithm was employed to efficiently compute time-origin averaged MSD over all valid lag times.

4.1 Power-law Scaling

The stable lag-time regime of each block was fitted using:

$$MSD(\tau) = A\tau^\alpha$$

where:

- A is the prefactor
- α is the diffusion exponent

Interpretation of α :

- $\alpha = 1$: Normal diffusion
- $\alpha < 1$: Subdiffusive motion

- $\alpha > 1$: Superdiffusive motion

Only the statistically stable early-time regime was used for fitting in order to avoid noisy long-lag artifacts.

4.2 MSD Fitting Parameters

Block	A	α
1	1.0149	0.3591
2	8.2868×10^{-1}	0.4107
3	6.7662×10^{-1}	0.5551
4	6.1139×10^{-1}	0.6015
5	4.5108×10^{-1}	0.4067
6	4.2879×10^{-1}	0.4450
7	3.0072×10^{-1}	0.4942

Table 2: Power-law fitting parameters obtained from blockwise MSD analysis

Most blocks exhibit subdiffusive behavior, indicating constrained polymer motion and viscoelastic relaxation effects.

4.3 MSD Plot

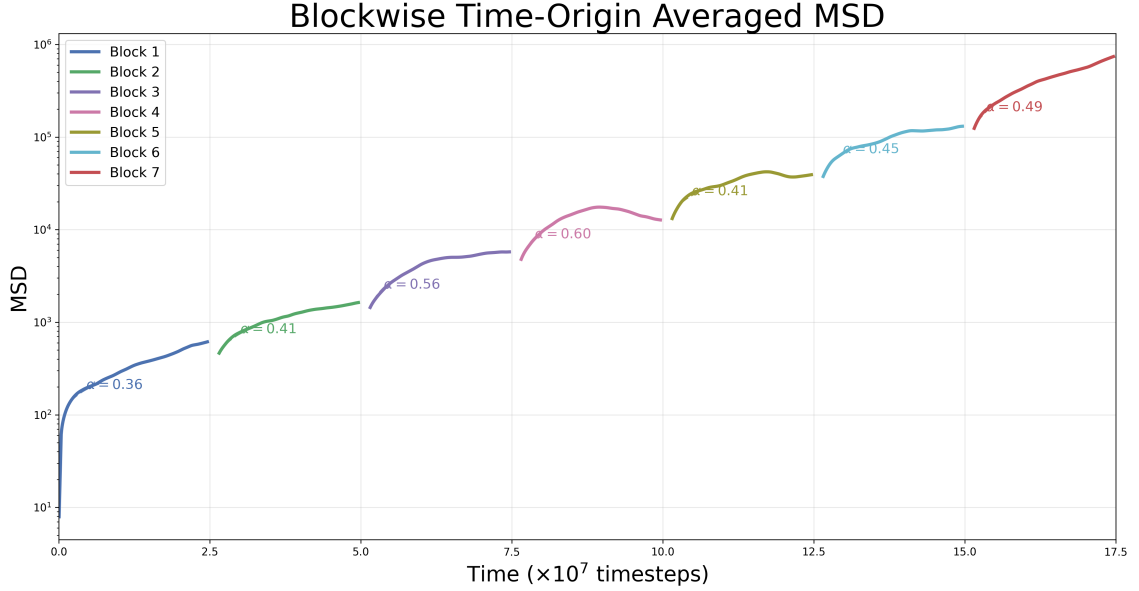


Figure 1: Blockwise MSD curves during temperature annealing

The MSD curves show temperature-dependent changes in polymer mobility and relaxation dynamics.

4.4 Discussion

The following observations were made:

- MSD increases continuously with lag time.
- Different annealing blocks exhibit different scaling exponents.
- Most blocks show subdiffusive behavior.
- Long-lag regions become statistically noisy due to reduced time-origin averaging.
- At large lag times, the number of available time origins decreases significantly, resulting in increased statistical noise in the MSD curves.
- Polymer mobility evolves significantly during annealing.
- Intermediate annealing stages show enhanced monomer mobility before final structural relaxation and compaction.

5 Radius of Gyration (R_g)

The Radius of Gyration describes the spatial extent of the polymer chain and is defined as:

$$R_g = \sqrt{\frac{1}{N} \sum_{i=1}^N (\mathbf{r}_i - \mathbf{r}_{cm})^2}$$

where:

- N is the number of monomers
- $\mathbf{r}_i$ is the position of monomer i
- $\mathbf{r}_{cm}$ is the center-of-mass position

5.1 Blockwise Radius of Gyration Results

Block	R_g	Standard Deviation
1	15.2032	± 1.5248
2	12.8645	± 1.4233
3	11.3212	± 1.4006
4	8.7162	± 0.6558
5	7.9530	± 0.1581
6	7.4845	± 0.3029
7	7.0173	± 0.0596

Table 3: Blockwise Radius of Gyration values during annealing

The Radius of Gyration decreases progressively across annealing blocks, indicating gradual polymer compaction and structural relaxation during cooling.

5.2 R_g Plot

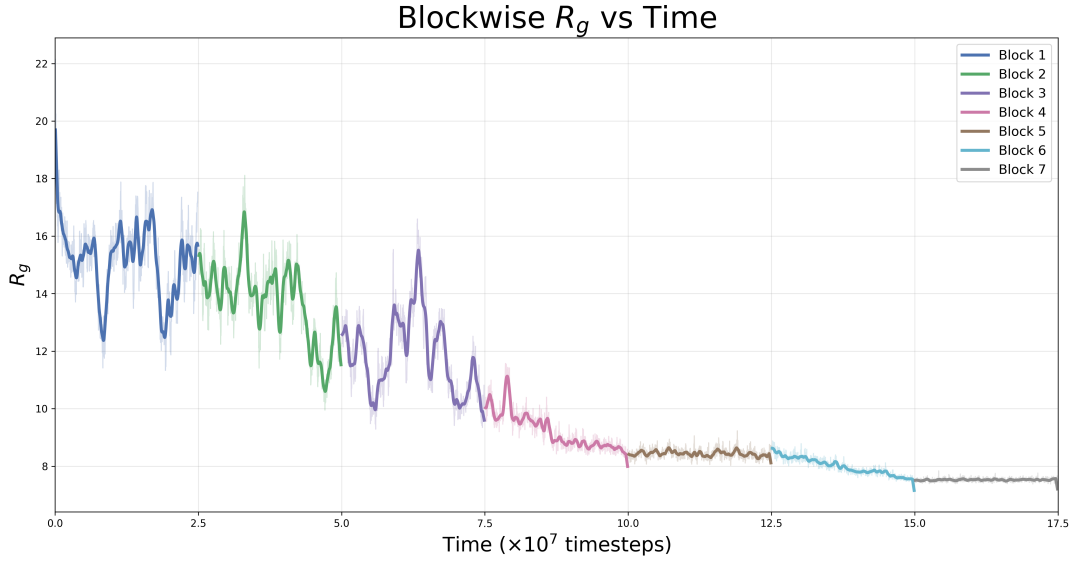


Figure 2: Blockwise Radius of Gyration during annealing

5.3 Discussion

The Radius of Gyration analysis reveals:

- progressive reduction in polymer size during annealing
- temperature-dependent conformational rearrangement
- increased structural compactness at later annealing stages

Fluctuations in R_g originate from thermal motion and configurational transitions of the polymer chain.

6 Conclusion

Blockwise analysis of polymer dynamics during temperature annealing was successfully performed using Molecular Dynamics trajectories generated from LAMMPS.

The study demonstrates:

- temperature-dependent polymer mobility
- dominant subdiffusive behavior in most annealing blocks
- progressive reduction in Radius of Gyration during cooling
- structural relaxation across annealing stages

The results highlight the importance of blockwise analysis for understanding polymer relaxation, transport properties, and conformational evolution during thermal annealing.

7 Software and Libraries Used

- Python
- NumPy
- Matplotlib
- FFT-based MSD computation
- LAMMPS trajectory files (.lammprj)